



Session Title: DFSMS Basics: DFSMSrmm Introduction for Storage Administrators

Session ID: 2403

Speaker Name: Mike Wood

NOTICES AND DISCLAIMERS

Copyright © 2009 by International Business Machines Corporation.

No part of this document may be reproduced or transmitted in any form without written permission from IBM Corporation.

Product information and data has been reviewed for accuracy as of the date of initial publication. Product information and data is subject to change without notice. This document could include technical inaccuracies or typographical errors. IBM may make improvements and/or changes in the product(s) and/or programs(s) described herein at any time without notice.

References in this document to IBM products, programs, or services does not imply that IBM intends to make such such products, programs or services available in all countries in which IBM operates or does business. Consult your local IBM representative or IBM Business Partner for information about the product and services available in your area.

Any reference to an IBM Program Product in this document is not intended to state or imply that only that program product may be used. Any functionally equivalent program, that does not infringe IBM's intellectual property rights, may be used instead. It is the user's responsibility to evaluate and verify the operation of any non-IBM product, program or service.

The information provided in this document is distributed "AS IS" without any warranty, either express or implied. IBM EXPRESSLY DISCLAIMS any warranties of merchantability, fitness for a particular purpose OR NONINFRINGEMENT. IBM shall have no responsibility to update this information. IBM products are warranted according to the terms and conditions of the agreements (e.g., IBM Customer Agreement, Statement of Limited Warranty, International Program License Agreement, etc.) under which they are provided. IBM is not responsible for the performance or interoperability of any non-IBM products discussed herein.

The performance data contained herein was obtained in a controlled, isolated environment. Actual results that may be obtained in other operating environments may vary significantly. While IBM has reviewed each item for accuracy in a specific situation, there is no guarantee that the same or similar results will be obtained elsewhere.

The responsibility for use of this information or the implementation of any of these techniques is a customer responsibility and depends on the customer's or user's ability to evaluate and integrate them into their operating environment. Customers or users attempting to adapt these techniques to their own environments do so at their own risk. IN NO EVENT SHALL IBM BE LIABLE FOR ANY DAMAGE ARISING FROM THE USE OF THIS INFORMATION, INCLUDING BUT NOT LIMITED TO, LOSS OF DATA, BUSINESS INTERRUPTION, LOSS OF PROFIT OR LOSS OF OPPORTUNITY.

Information concerning non-IBM products was obtained from the suppliers of those products, their published announcements or other publicly available sources. IBM has not necessarily tested those products in connection with this publication and cannot confirm the accuracy of performance, compatibility or any other claims related to non-IBM products. Questions on the capabilities of non-IBM products should be addressed to the suppliers of those products.

The provision of the information contained herein is not intended to, and does not, grant any right or license under any IBM patents or copyrights. Inquiries regarding patent or copyright licenses should be made, in writing, to:

IBM Director of Licensing
IBM Corporation
North Castle Drive
Armonk, NY 10504-1785
USA

Trademarks

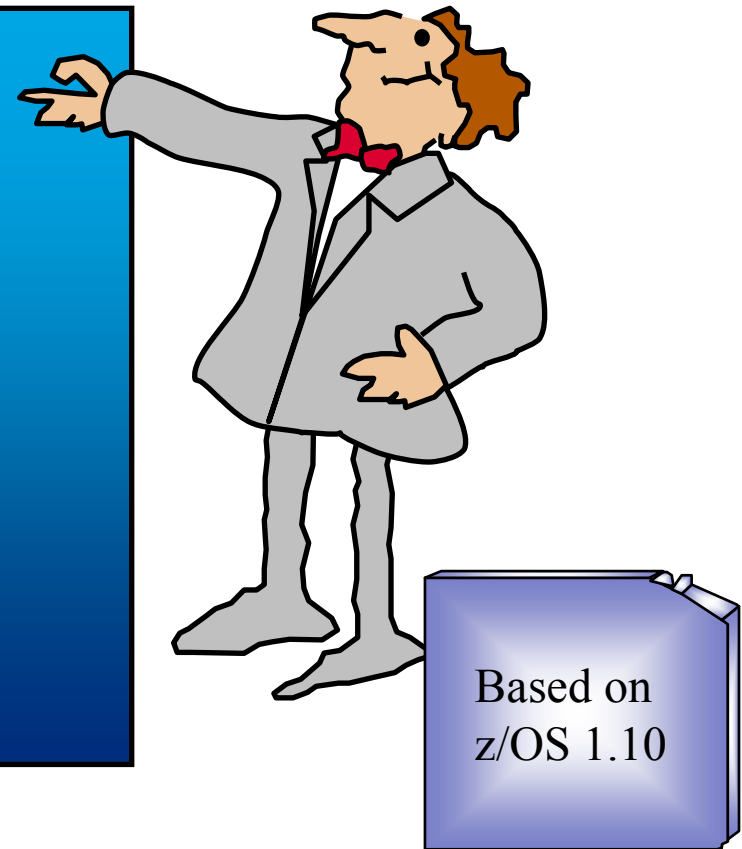
The following terms are trademarks of International Business Machines Corporation in the United States, and/or other countries:

- IBM
- DFSMSrmm
- IMS
- MVS
- RACF
- z/OS

Other company, product, and service names may be trademark or service marks of others.

Agenda

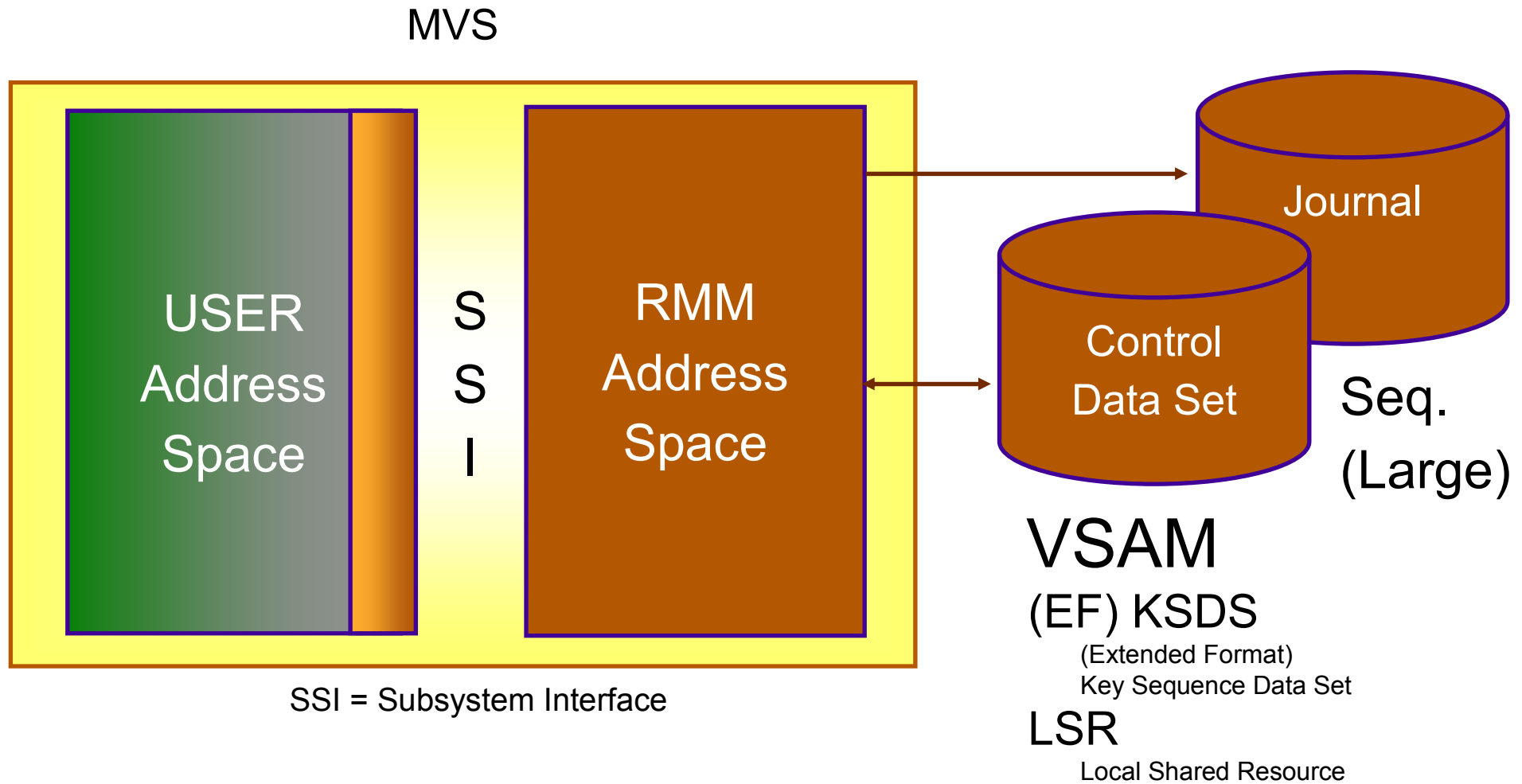
- Introduction to RMM
- Resources managed by RMM
- Software Interfaces
- User/Program Interfaces
- Inventory Management
- Tools



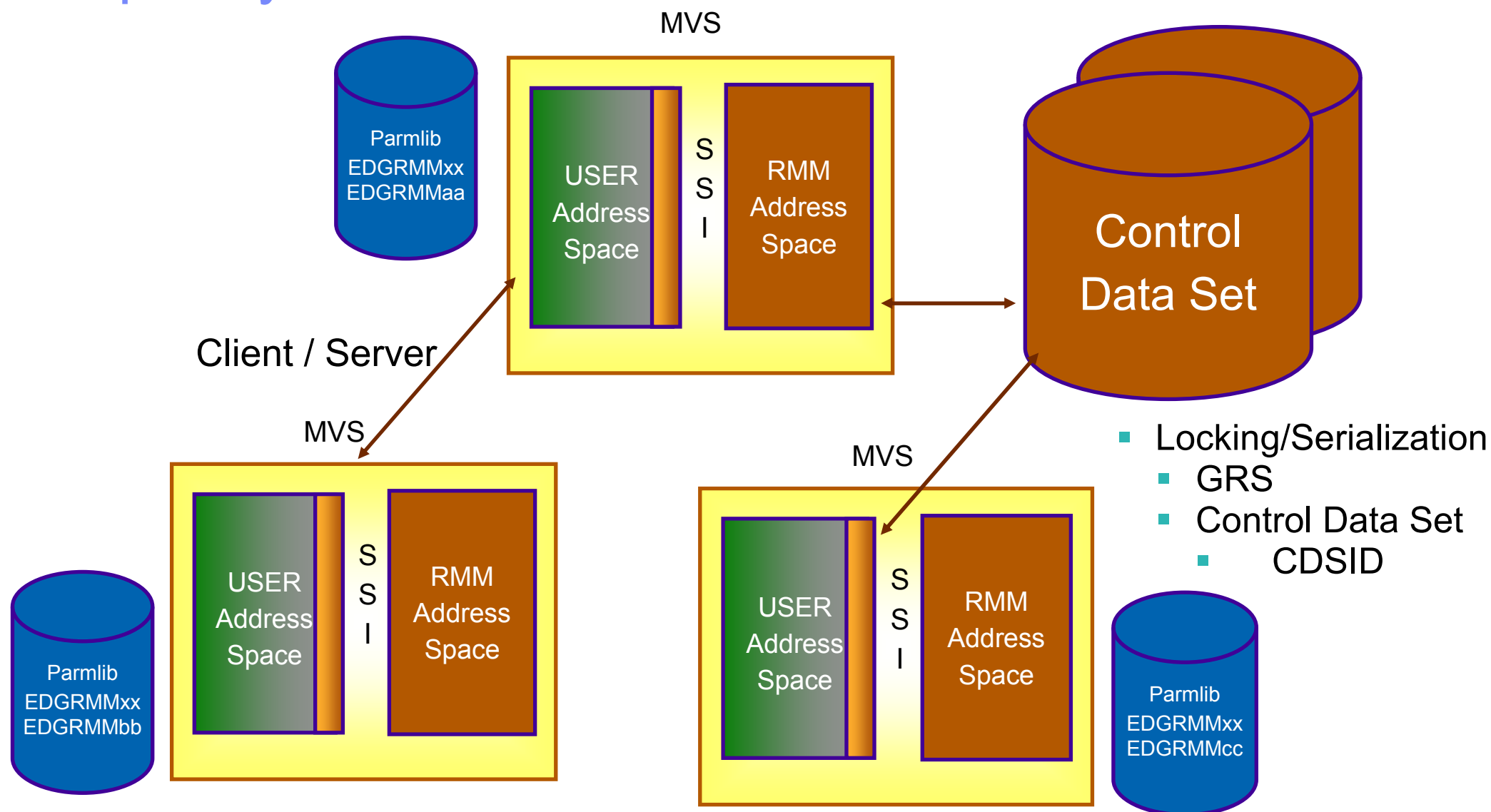
our product ...DFSMSrmm

- Removable Media Manager
 - IBM's tape management system for z/OS
 - A complete, centralized and automated system
 - Supports all tape devices and tape libraries
- Protecting tape data sets from accidental destruction
- Manage volume retention and movement
- Support and enhance system-managed tapes
- An Element of z/OS

Structure...

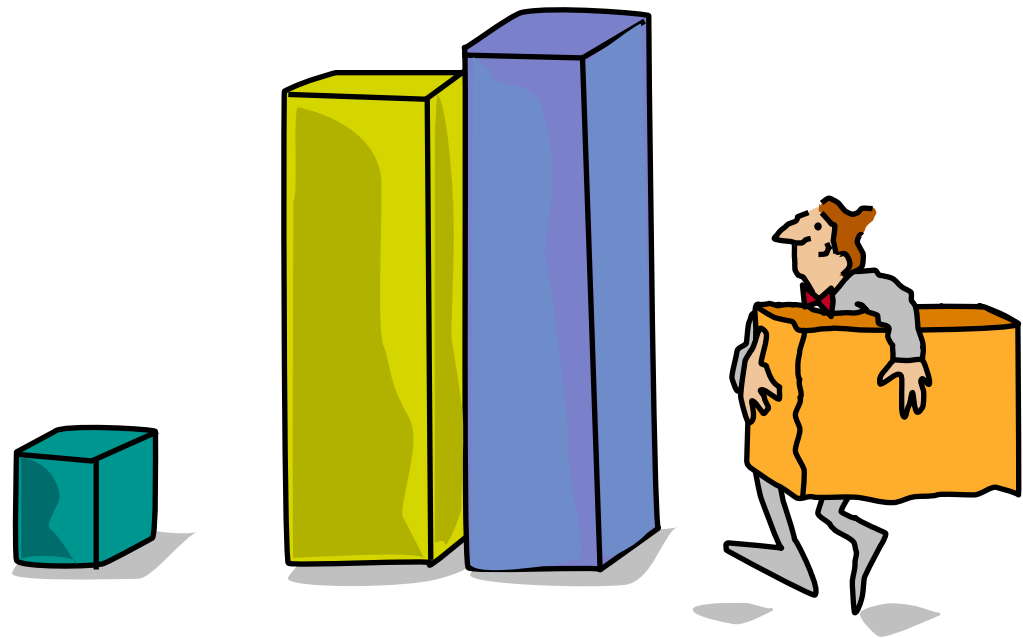


Multiple Systems ...

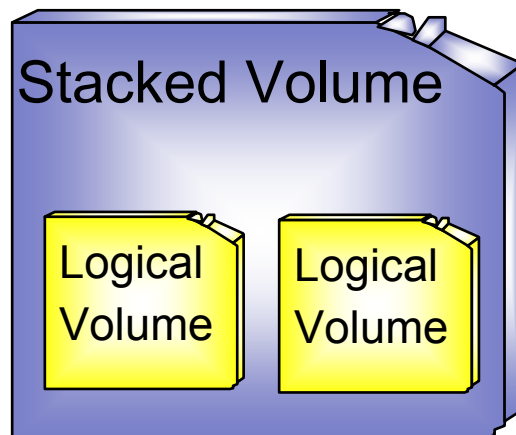
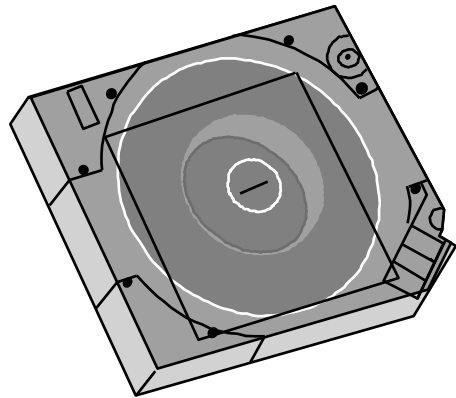


RMM managed resources

- Volume
- Data Set
- Owner
- Location
- Rack / Bin
- Vital Record Specification
- Software Product



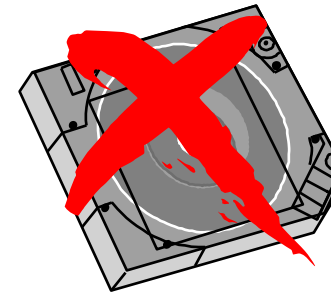
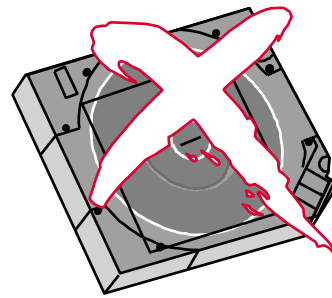
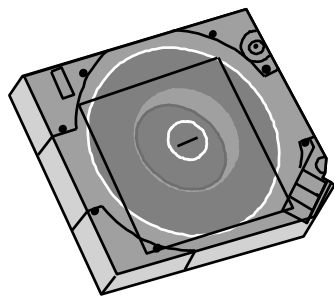
Volume ... Types



- Physical Volume
 - WORM WWUCID
- Logical Volume
 - Used in a Virtual Tape Server
 - Logical Volume / Virtual Volume
- Stacked Volume
 - Container for exported Logical Volumes (VTS)
 - Container for Copy Export
 - Box for Physical Volumes

Volume ...

- Volume Status
 - Master
 - Scratch
 - User
- Expiration Date
- Availability
 - Pending Release
 - Vital Record retained
 - On Loan
 - Open
- Retention Date



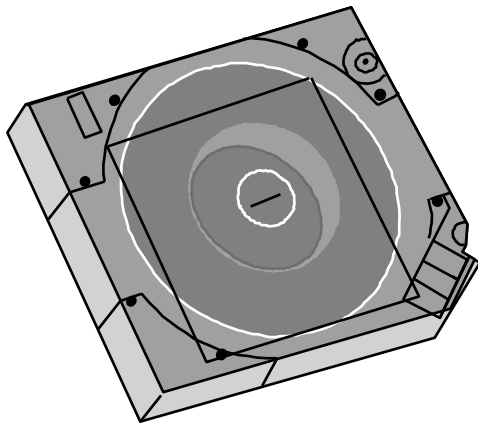
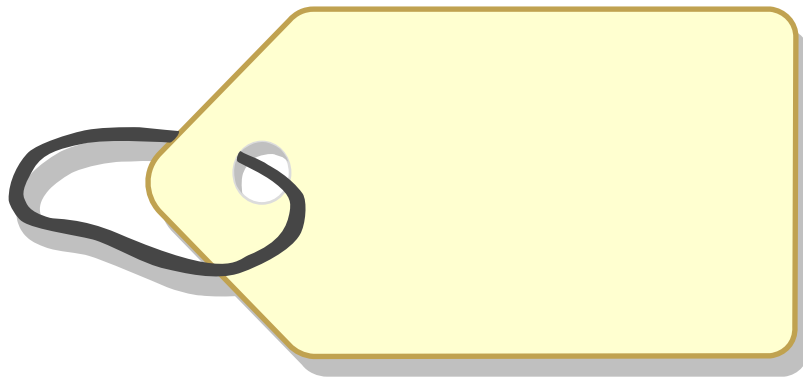
Volume ...

Validation

- Non specific mount request
 - Only scratch volumes can be mounted
- Specific mount request
 - The correct, non-scratch volume must be mounted
- PRTITION and OPENRULE commands in parmlib
 - Subset volumes and processing options
- When a data set is opened and closed DFSMSrmm automatically updates the volume and data set record



Volume ...



- Label
 - Standard Label
 - ANSI Label
 - No-Label

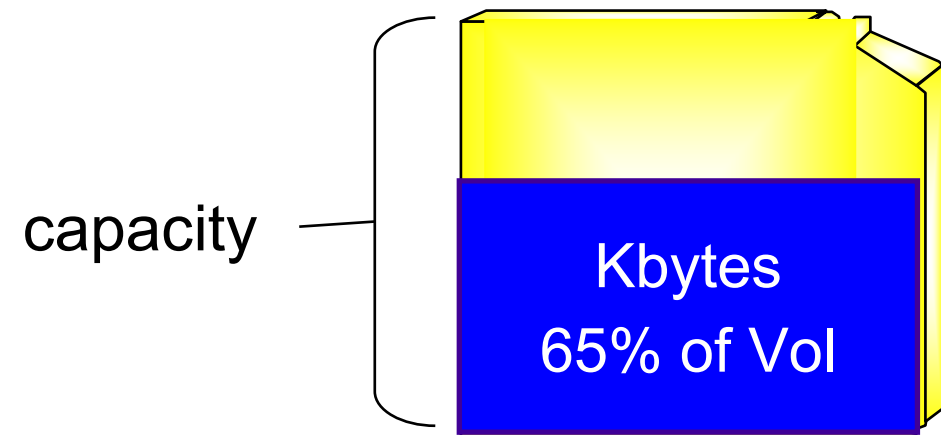
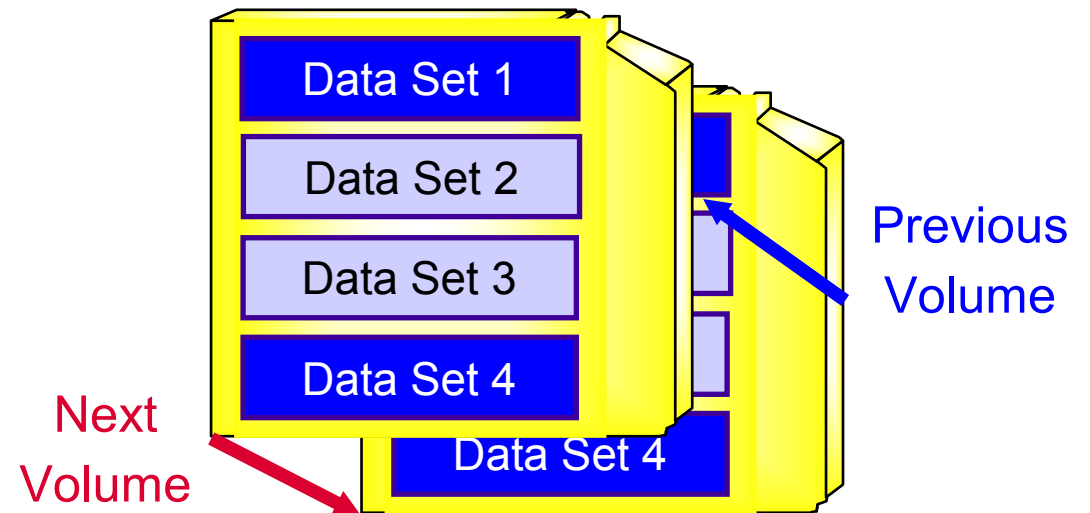
**duplicate
VOL1
supported**

- Media Attributes
 - Type
 - Capacity
 - Recording format
 - Compaction

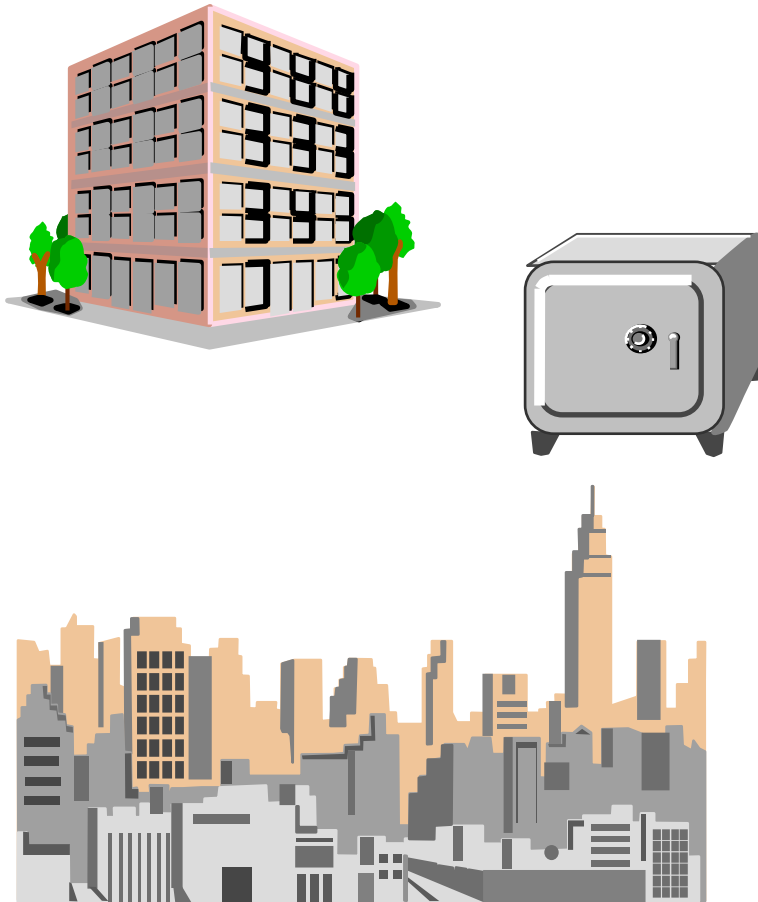
**MEDINF
for non-
IBM media**

Volume ...

- Data Sets on Volume
 - First DSN
 - Number of data sets
- Multi Volume Set
 - Next Volume
 - Previous Volume
 - Volume Sequence
- Volume usage
 - Application written
 - Hardware recorded



Volume ... Location



- Current Location
- Home Location
- Required Location
- Destination Location
- Old Location
- Loan Location
- Rack / Bin
- In Transit Status

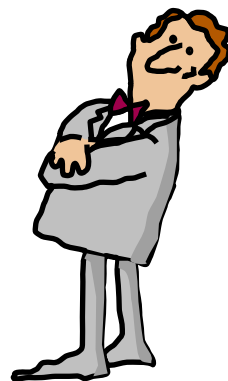
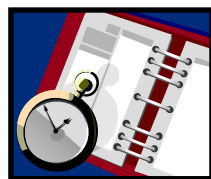
Volume ... Tracking

- Last Change

- Date/Time

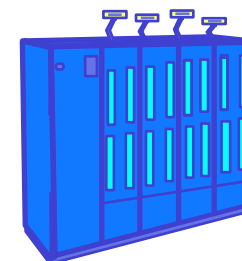
- Userid

- Systemid

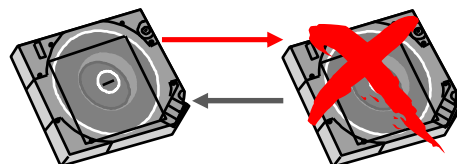


- Last Read Date

- Last Write Date



- Status Change Date/Time



- Drive last used

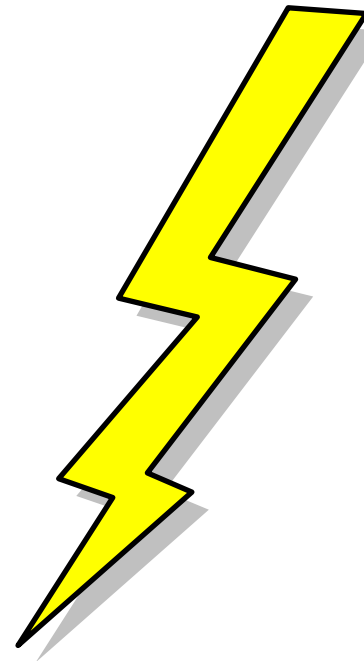
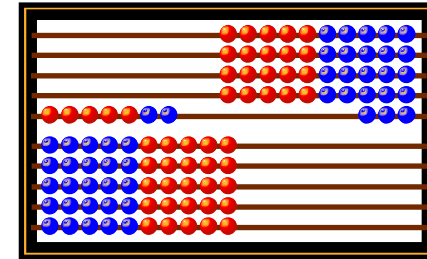
- Last Move Date

- Write Mount Count

Volume ... Tracking

- Volume use count
- Write Mount count

- Errors
 - Temporary
 - Read
 - Write
 - Permanent
 - Read
 - Write



Volume ... Release Actions

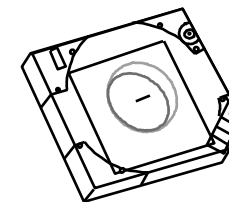
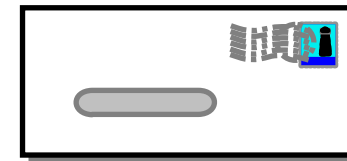
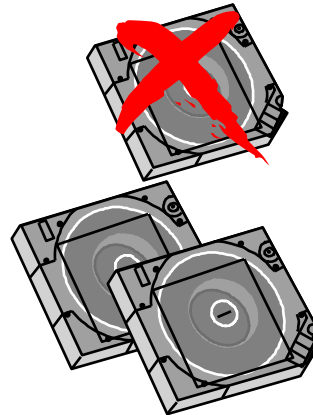
- **Return to Scratch**

- **Replace**

- **Return to Owner**

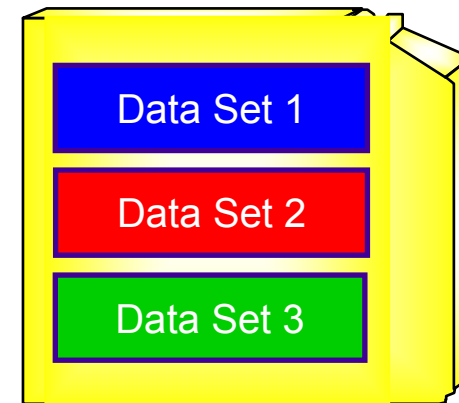
- **Notify Owner**

- **Init / Erase**

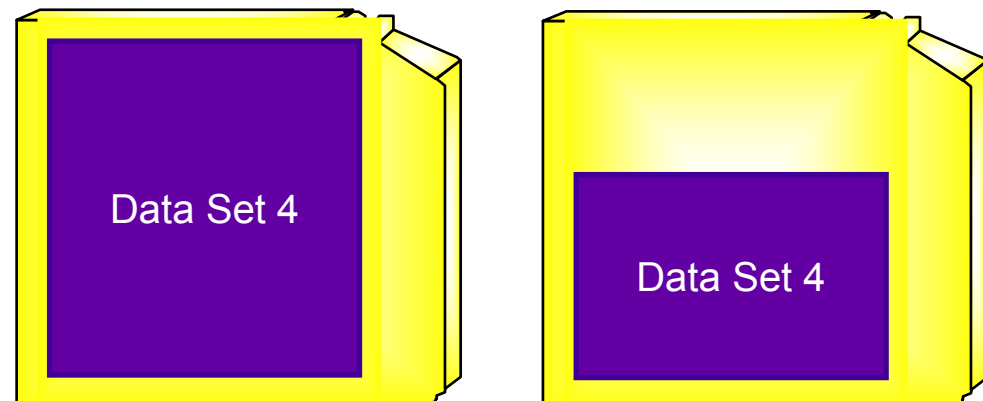


Data Set ...

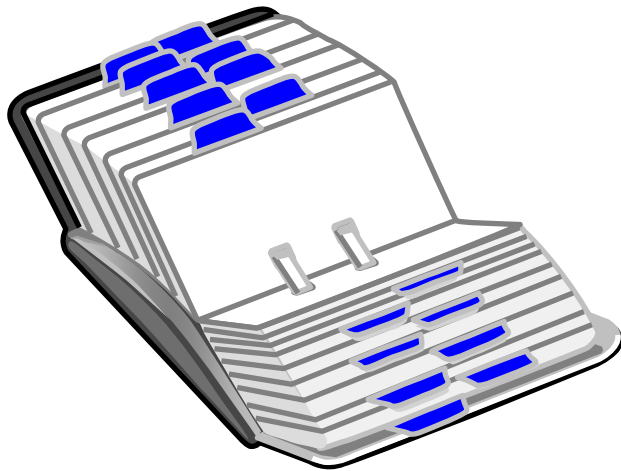
- Multiple Data Sets on one Volume



- One Data Set on multiple Volumes



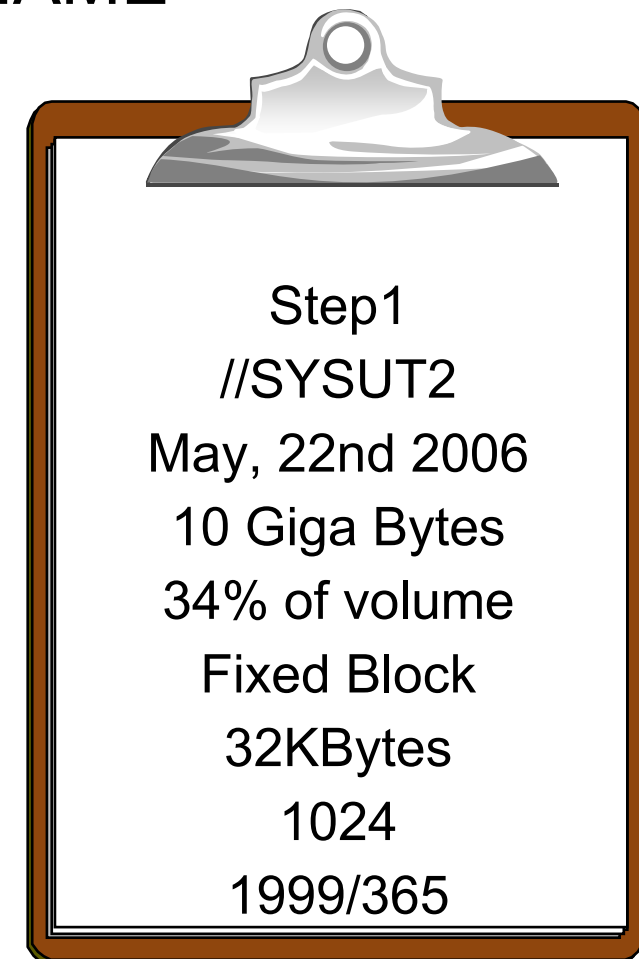
Data Set ... Control Info



- Key
 - Data Set Name (44 Character)
 - Volume Serial
 - Physical File Sequence Number
- Job Name
- Management Class
- Management Value

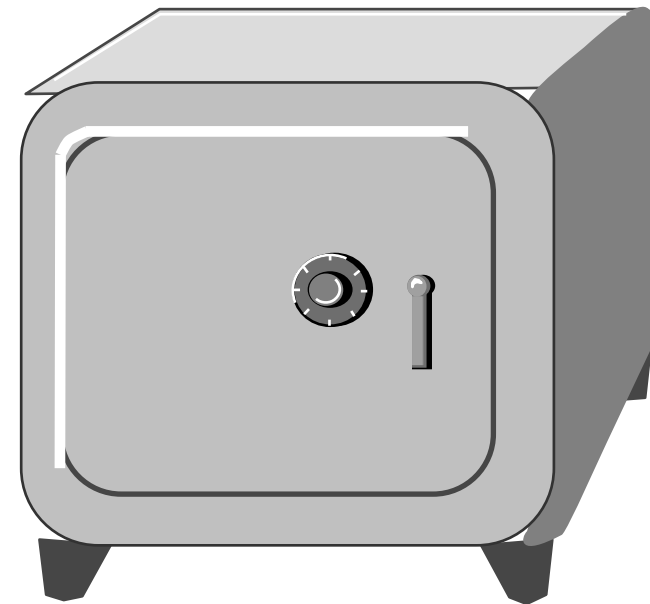
Data Set ... Attributes

- Step Name / Program Name / DD NAME
- Create Date / Time
- Date last read / written
- Data Set Size
- Record Format
- Block Size
- Logical Record Length
- EXPDT and Original EXPDT
- ABEND, DELETED



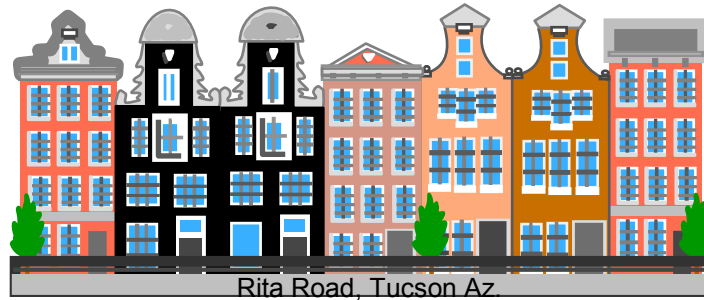
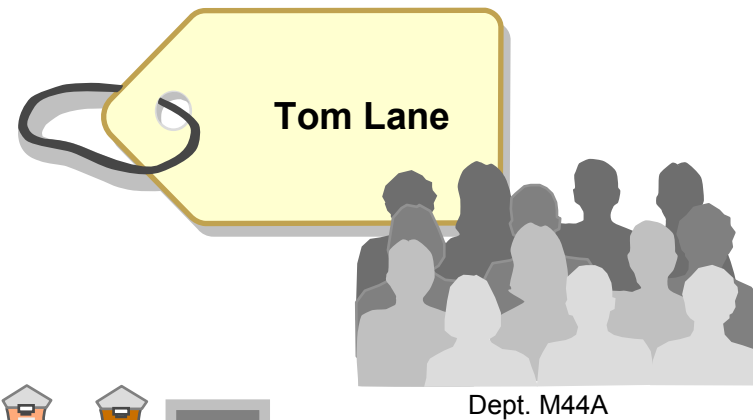
Data Set ... Vital Record

- Vital Record Status
- Matching Vital Record Specification
- Current VRS Sub-Chain
 - name
 - date
- Retention Date

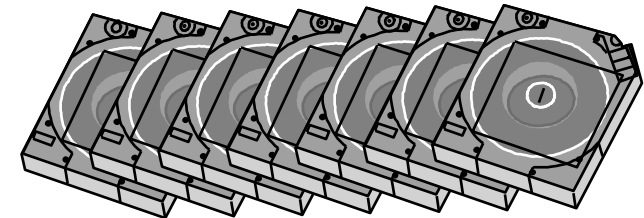


Owner ...

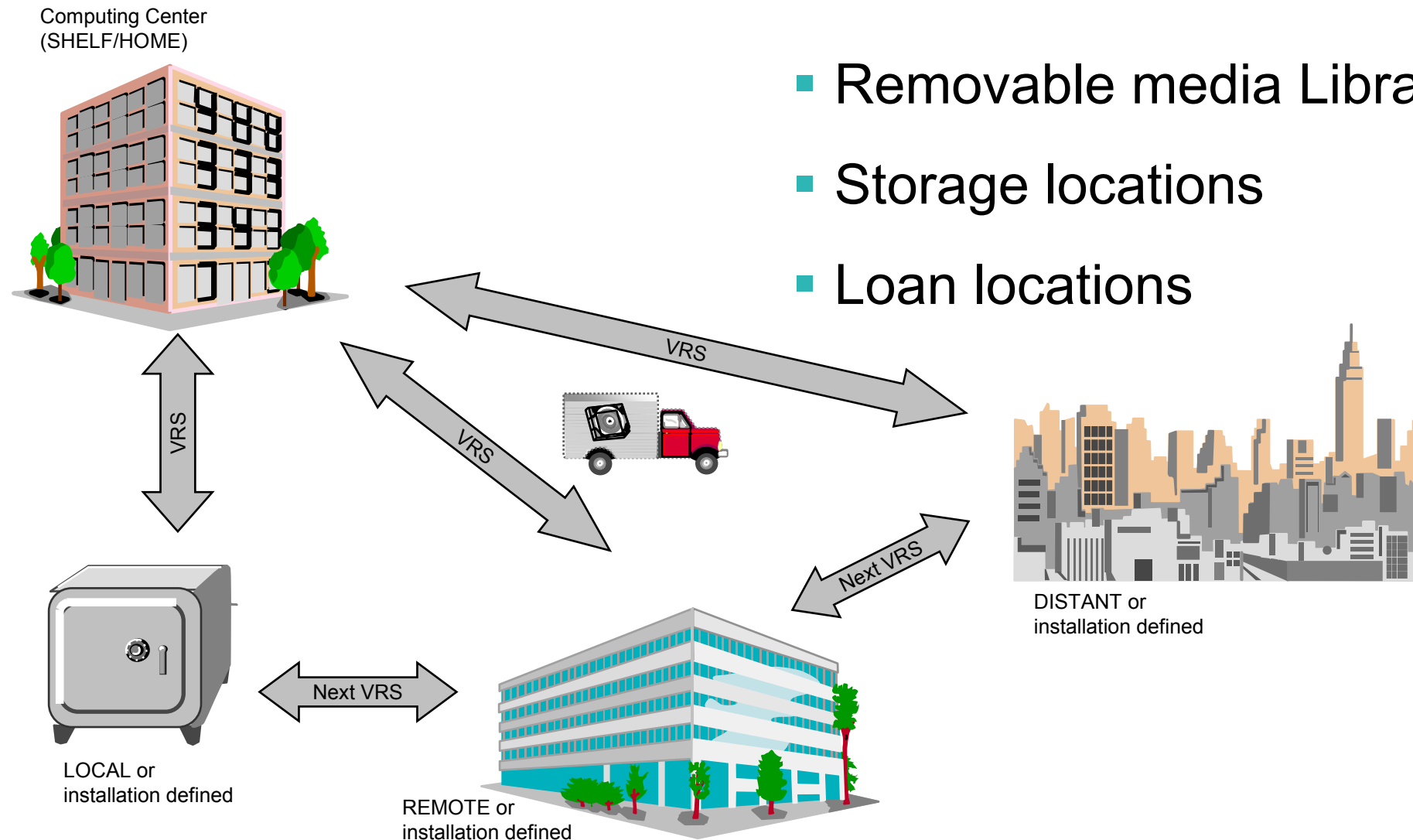
- Name, Department, etc.
- Address
- E-mail
 - user@hostname or Node.Userid



- List of owned Volumes



Location...



Vital Record Specification

- VRS's are Policies to ...
 - control Retention of Data Sets and Volumes
 - determine Volume's required Location
 - Set Release Options

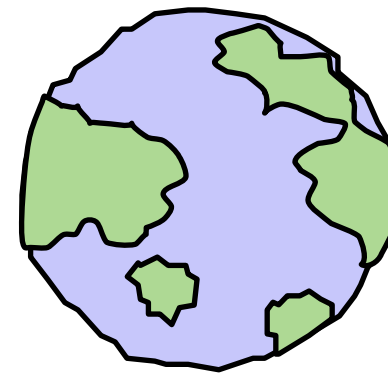


- **VRSEL**

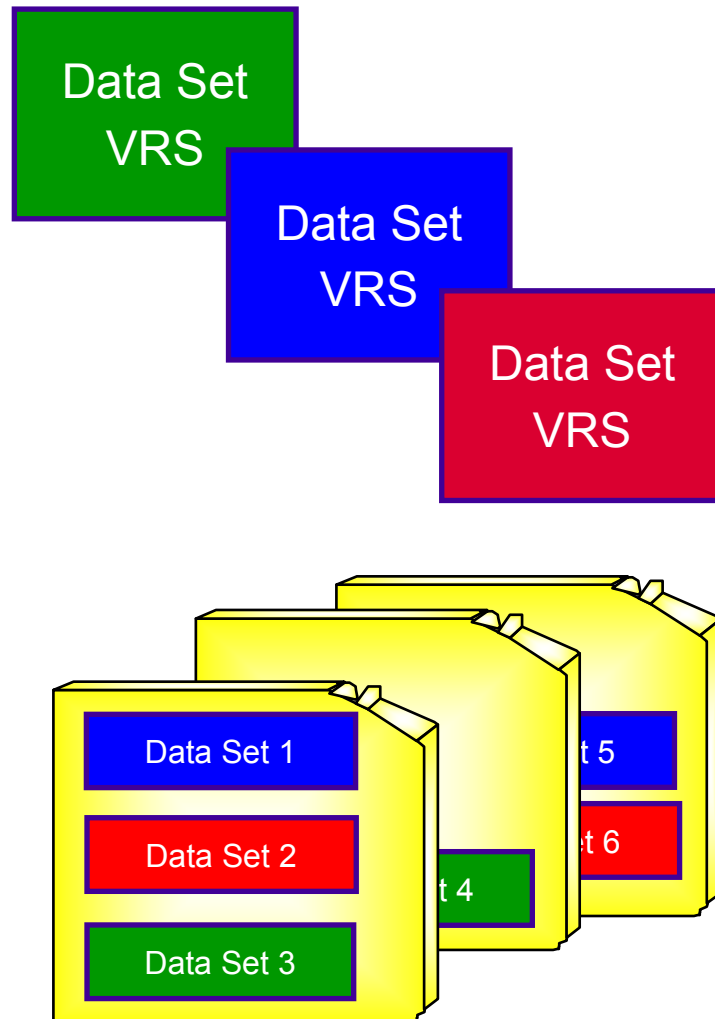
- NEW



- OLD

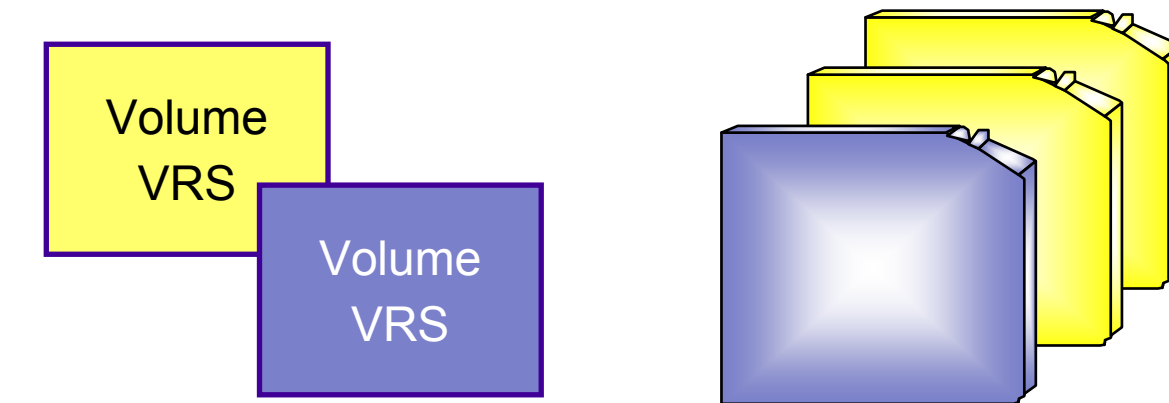


Vital Record Specification

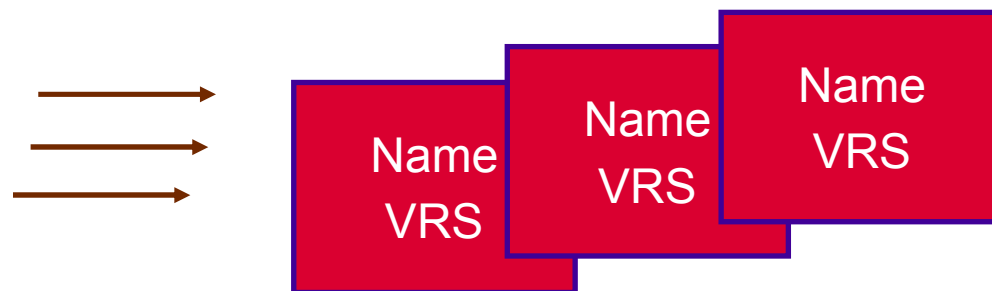


- Data Set VRS
 - Select Data Sets by:
 - OPEN flag
 - DELETED flag
 - ABEND flag
 - Management Class
 - Management Value
 - Data Set Name
 - Jobname
 - Generation Data Group
 - Generic Masks supported
 - Plus - Pseudo GDG

Vital Record Specification



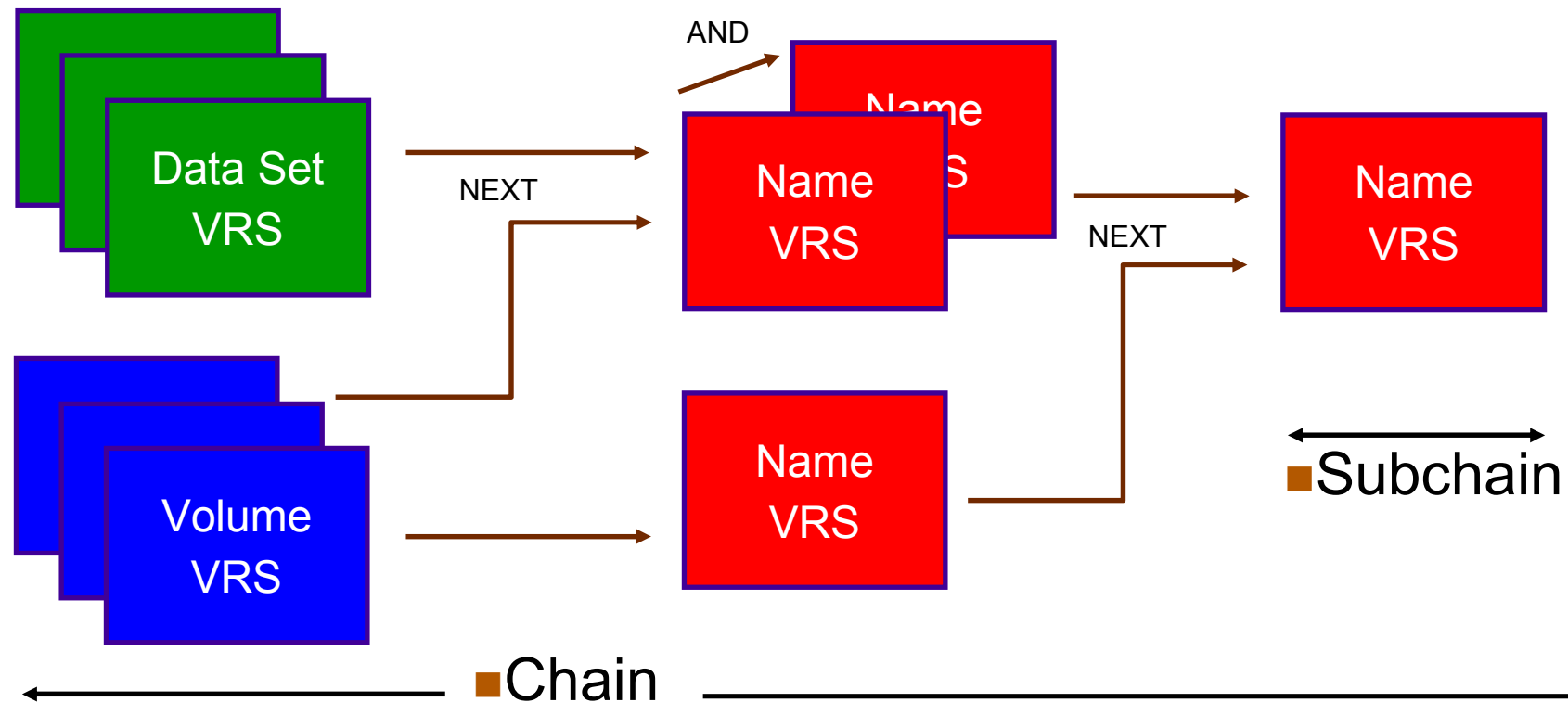
- Volume VRS
 - Specific
 - Generic



- Name VRS's
 - Location only
 - Retention & Location

Vital Record Specification

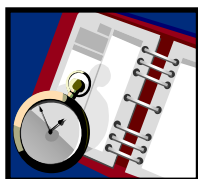
- Data Set VRS
- Volume VRS
- Name VRS



Vital Record Specification

Retention Types

- Specific Date



- Cycles

- By Days Cycle

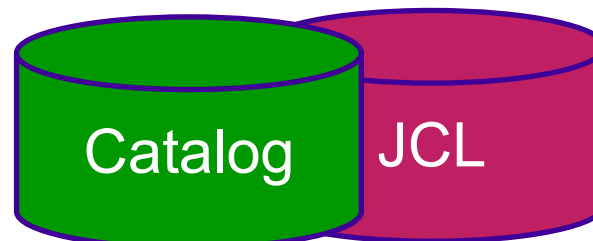


- While Cataloged

- Until Expired

- Days since creation
- Days since last reference
- Extra days

- Forever

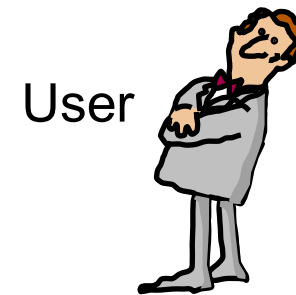


Vital Record Specification

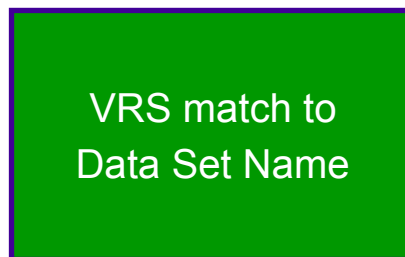


Administrator

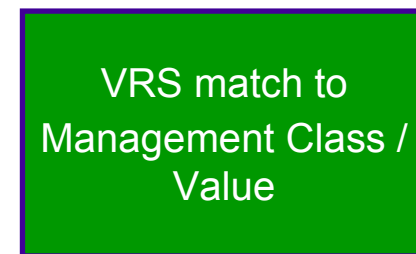
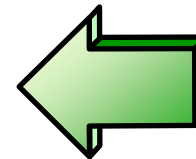
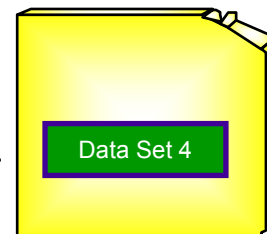
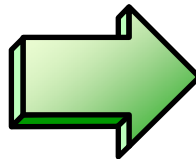
- Option - Two Policies manage one data set



User



Primary VRS



Secondary VRS



Location



Retention

Audit Controls for VRS and Release Processing

- **Avoiding Loss of Data**

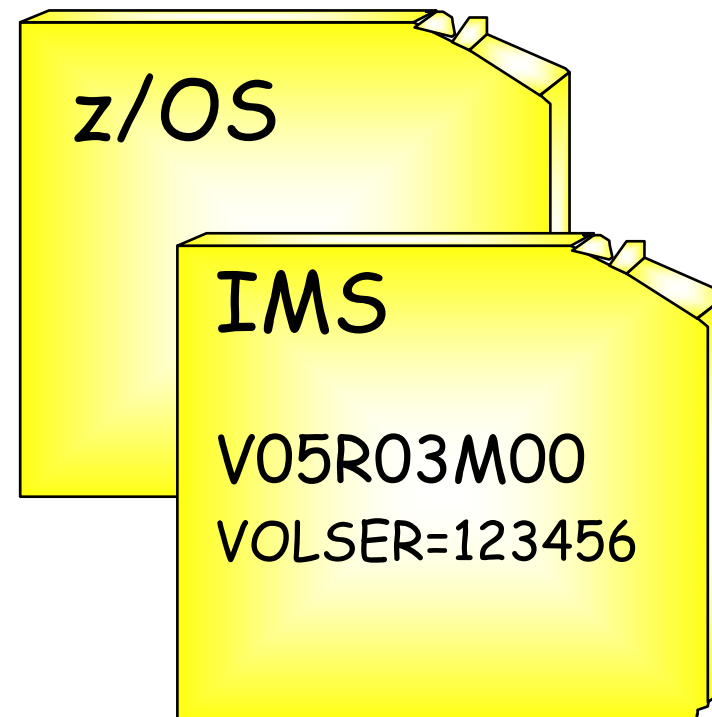
- Pre-empt system errors, user errors
- Define expected retention as an overall control
- Exploit automated way, to detect when too many volumes are released or expired

- **New parmlib options are added**

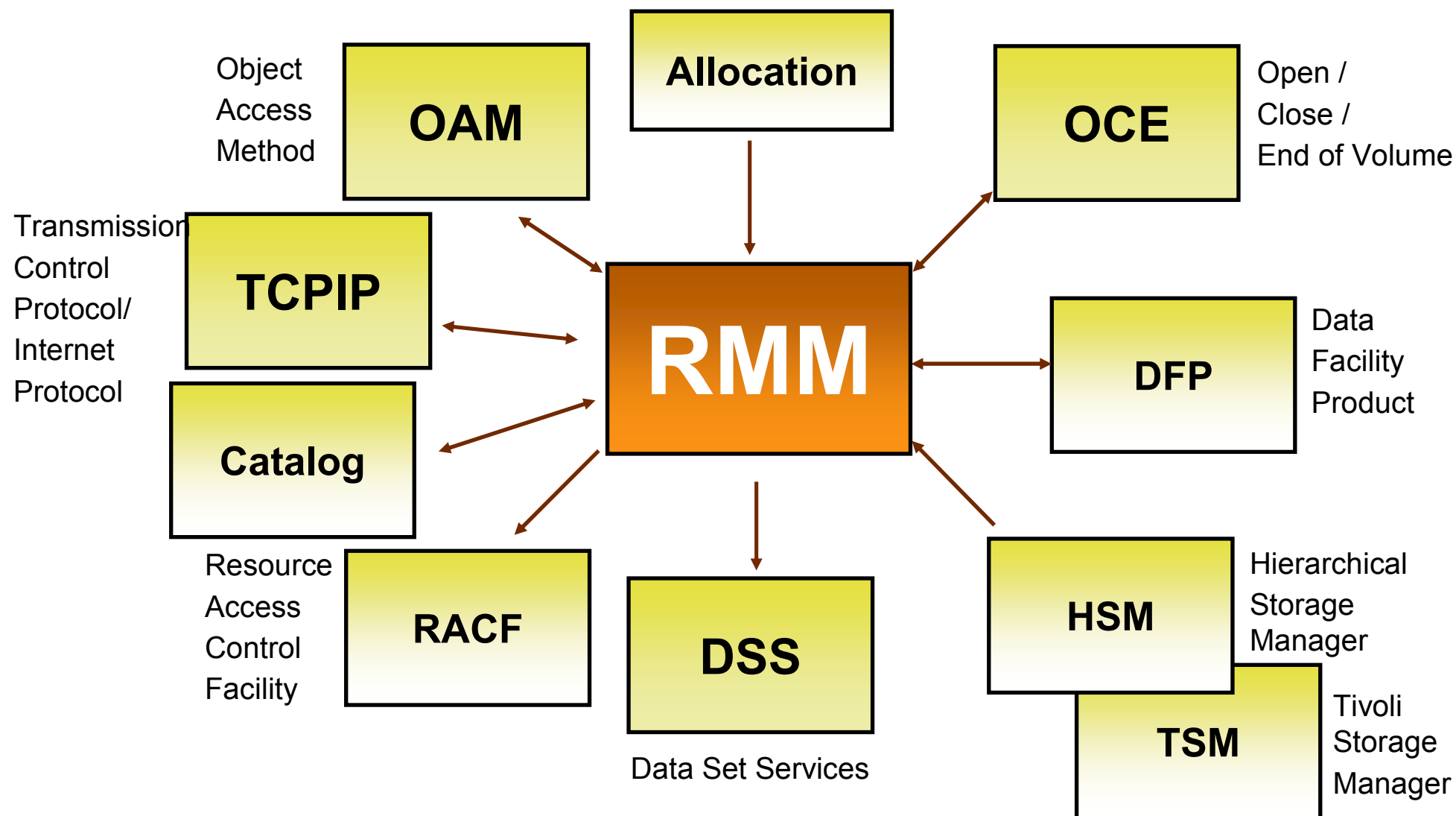
- Use VRSDROP to control volumes dropped from VRS retention
- Use VRSRETAIN to control newly assigned volumes that must be retained by VRS
- Use EXPDTDROP to control volumes dropped from EXPDT retention

RMM managed resources

- Volume
- Data Set
- Owner
- Location
- Rack / Bin
- Vital Record
- Specification
- **Software Product**

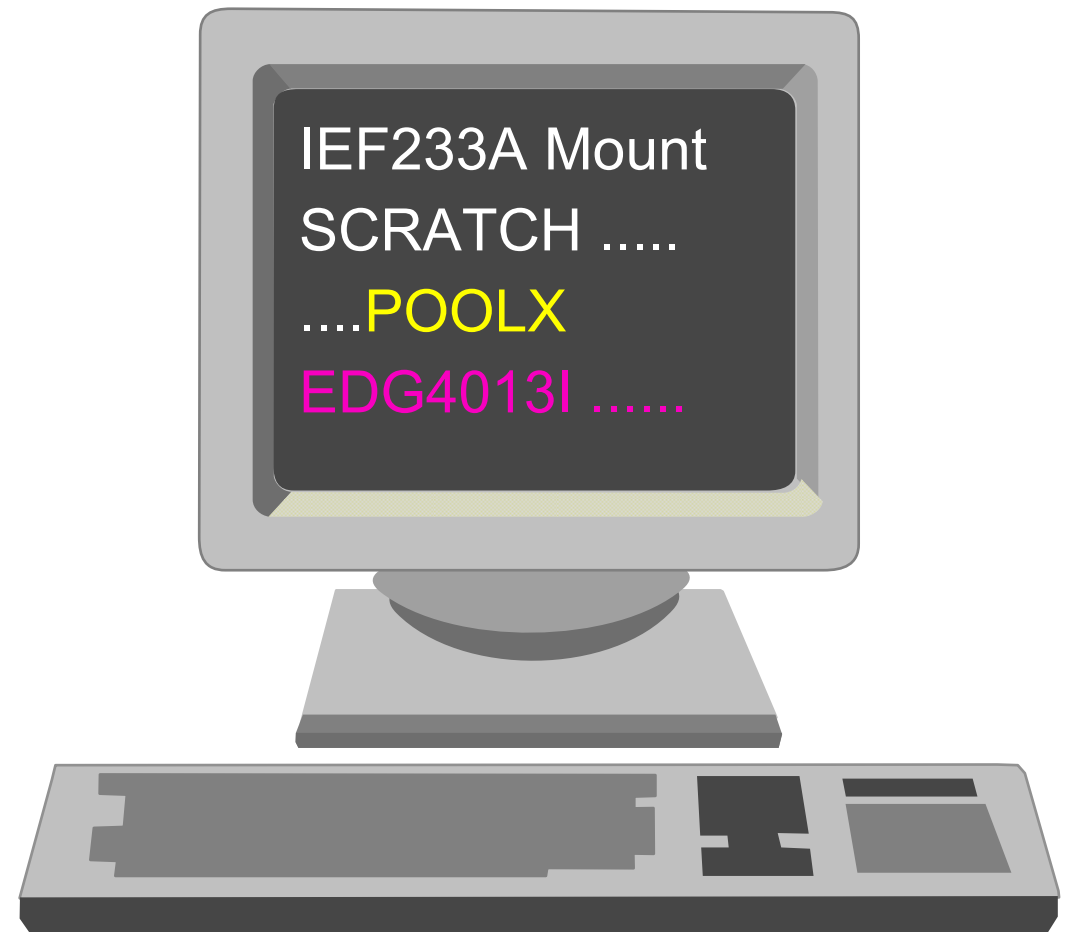


Software interfaces



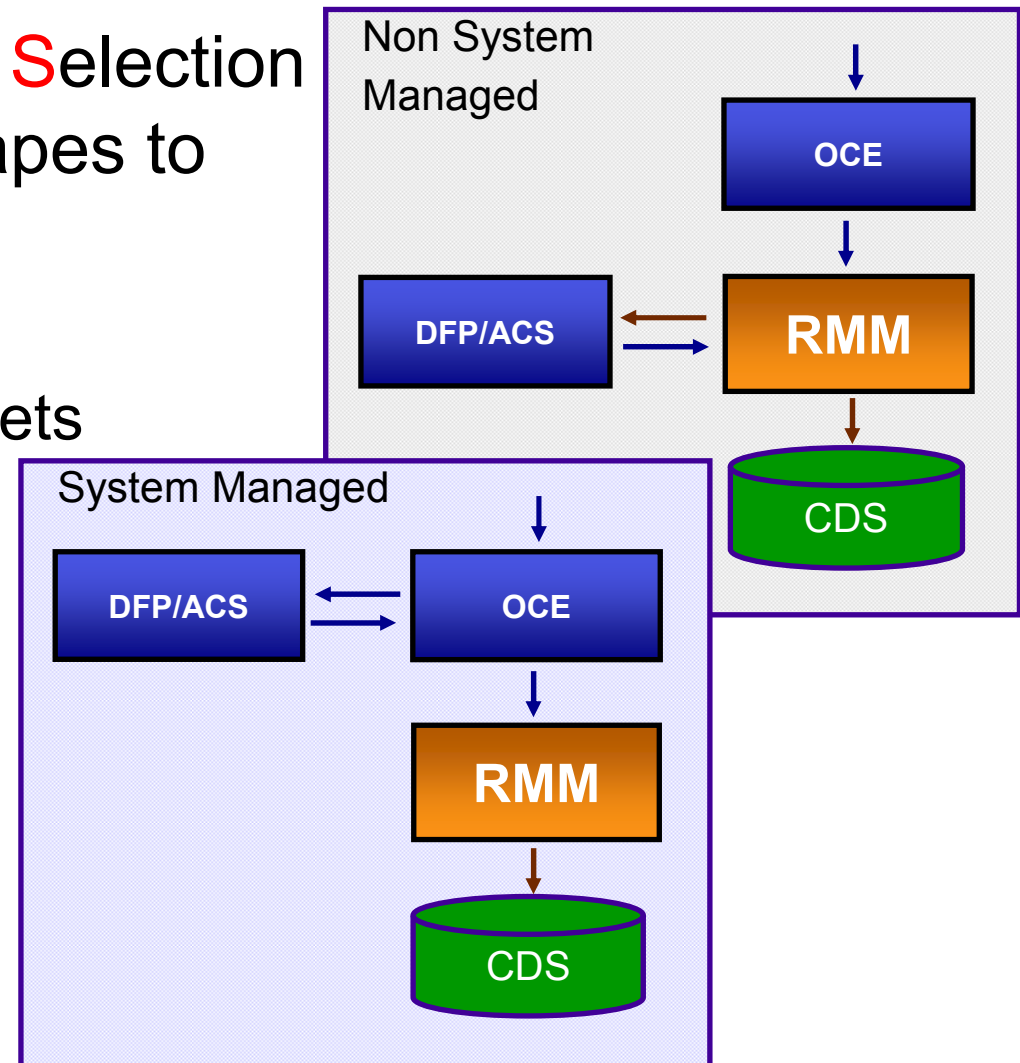
Allocation

- Intercept Mount Message
- Assign Scratch Pool /
- Find Rack
- Modify Mount Message
- EDG4013I if unavailable



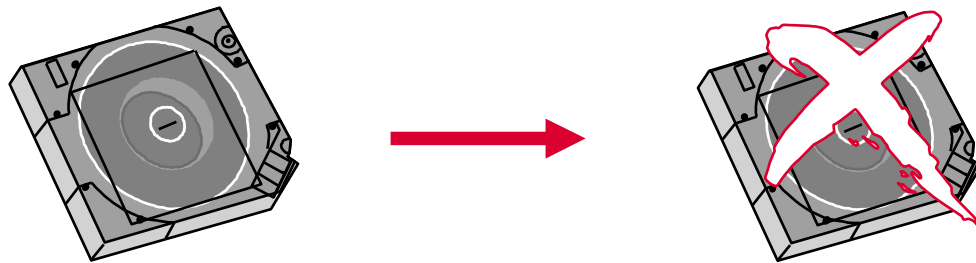
Data Facility Product

- RMM calls **A**utomated **C**lass **S**election for Non System-Managed Tapes to get:
 - Storage Group for Volumes
 - Management Class for Data Sets
- DEVSUPxx Tape Security
 - TAPEAUTHDSN
 - TAPEAUTHF1
 - Erase on Scratch



Hierarchical Storage Manager, OAM, and Tivoli Storage Manager

HSM, OAM, and TSM call RMM to release a Volume



API - EDGTVEXT

Privileges for Volumes created by HSM

- only check 17 Character Data Set Name
 - for input processing

RACF

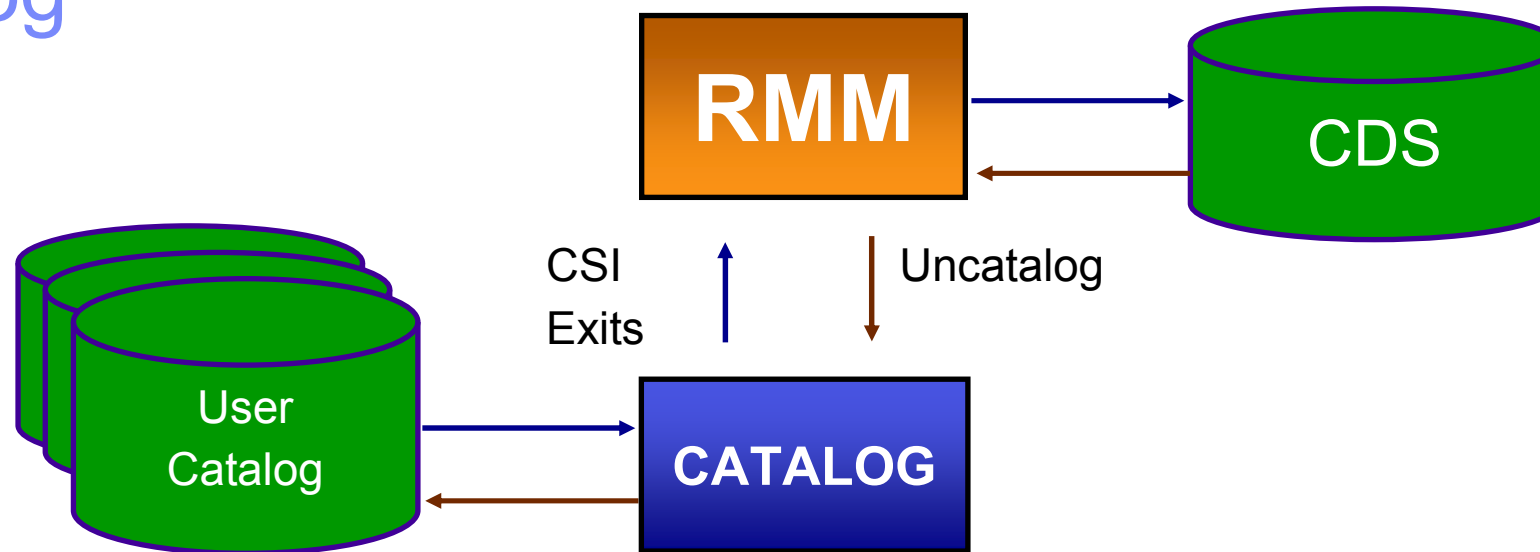
- Protect RMM Resources
- Authorize RMM Functions
 - 'All powerful' user
 - Granular Security option

**FACILITY:
STGADMIN.EDG.etc.**

- Erase on Scratch sets ERASE action
- Create and delete Tape Profiles
 - TAPEVOL
 - TAPEDSN
- Supports DEVSUPxx TAPEAUTHDSN+



Catalog



- Uses Catalog Search Interface to synchronize CDS with Catalog (tape data sets only)
- Exploits Catalog Exits to mirror Catalog updates in CDS
- Uncatalogs Tape Data Sets

User interfaces



↔ TSO Commands



RMM ADDVOLUME ...
RMM LISTDATASET ...
RMM SEARCHVOLUME
RMM CHANGEOWNER ...
RMM DELETEVRS ...

↔ ISPF Dialogs



DFSMSrmm User Menu

```
Option ==>
0  OPTIONS - Specify dialog options and defaults
1  VOLUME  - Display list of volumes
2  DATA SET - Display list of data sets
3  PRODUCTS - Display list of Products
4  OWNER   - Display or Change owner information
5  REQUEST - Request a new volume
6  RELEASE - Release an owned volume
R  REPORT  - Work with reports
```

ISPF Dialogs

REMOVABLE MEDIA MANAGER (DFSMSrmm)

Option ==>

- | | | |
|---|---------------|---------------------------------------|
| 0 | OPTIONS | - Specify dialog options and defaults |
| 1 | USER | - General user facilities |
| 2 | LIBRARIAN | - Librarian functions |
| 3 | ADMINISTRATOR | - Administrator functions |
| 4 | SUPPORT | - System support facilities |
| 5 | COMMANDS | - Full DFSMSrmm structured dialog |
| 6 | LOCAL | - Installation defined dialog |
| X | EXIT | - Exit DFSMSrmm Dialog |

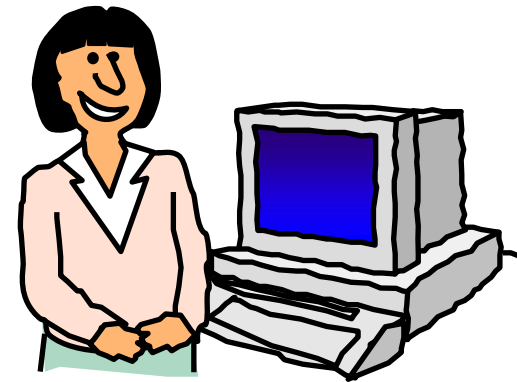
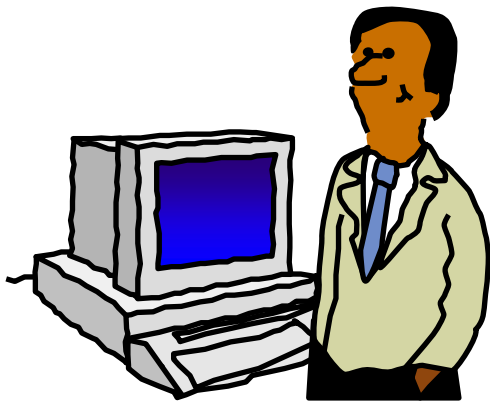
Enter selected option or END command.

For more info., enter HELP or PF1.

ISPF Dialogs

General User

- Request a scratch volume
- Change and release owned volumes / data sets
- Display most resources



Tape Librarian

- Define new volumes
- Add shelf locations
- Confirm volume movements and actions
- Report Generation

ISPF Dialogs

Storage Administrator

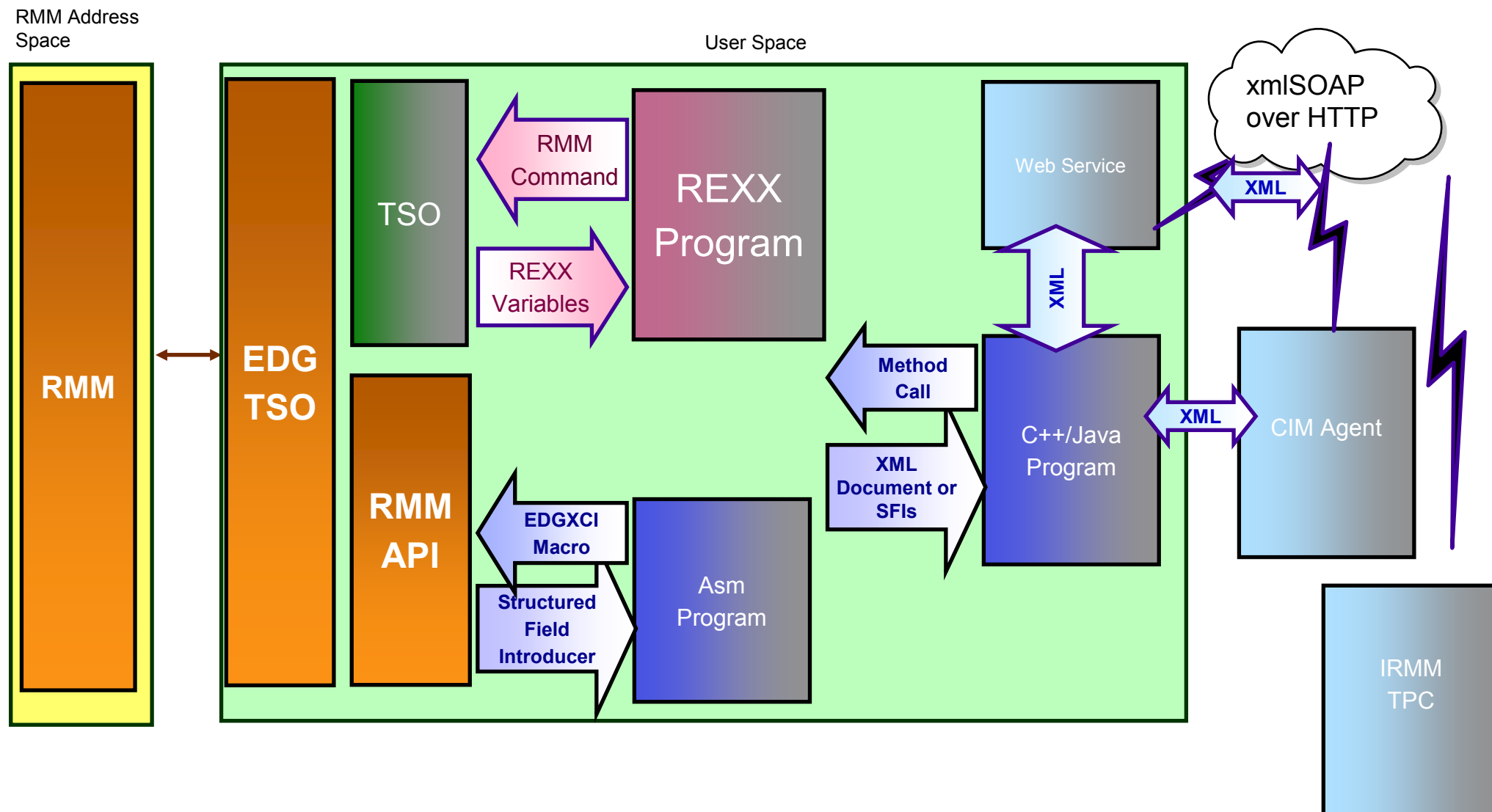
- Add, Change and Delete Vital Record Specifications
- Report Generation



System Programmer

- Maintain parmlib options and CDS control information
- List software products

Programming interfaces



Inventory Management

- Vital Record Processing
- Catalog Synchronization
- Expiration Processing
- Storage Location Mgmt.
- Report Extract
- Reports
- Backup



(Housekeeping)

Inventory Management

- **Vital Record Selection**
- Catalog Synchronization
- Expiration Processing
- Storage Location Mgmt.
- Report Extract
- Reports
- Backup

Retention

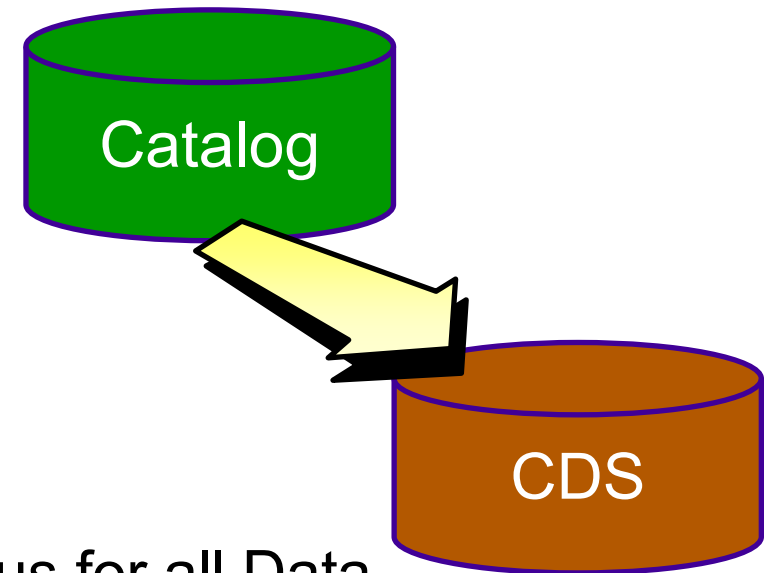


Location

- Apply VRS Policies to
 - Data sets
 - Volumes
 - Volume Sets
 - Stacked Volumes
- Apply Retention Limit Controls
- Vital Record Report
 - Unused VRS Report
- ACTIVITY File
- VERIFY run

Inventory Management

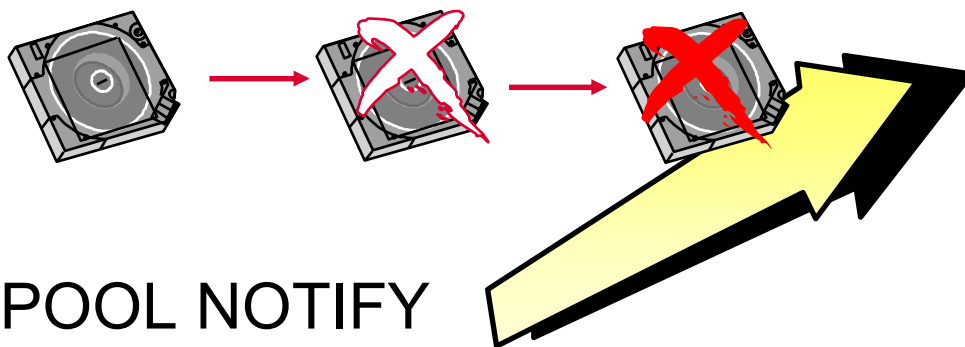
- Vital Record Selection
- **Catalog Synchronization**
- Expiration Processing
- Storage Location Mgmt.
- Report Extract
- Reports
- Backup



- Set Catalog Status for all Data Sets recorded in the CDS
- VERIFY
 - Report on Differences
- Run once to Implement

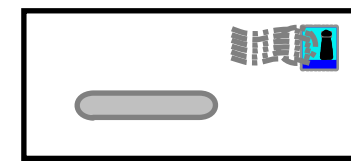
Inventory Management

- Vital Record Selection
- Catalog Synchronization
- **Expiration Processing**
- Storage Location Mgmt.
- Report Extract
- Reports
- Backup



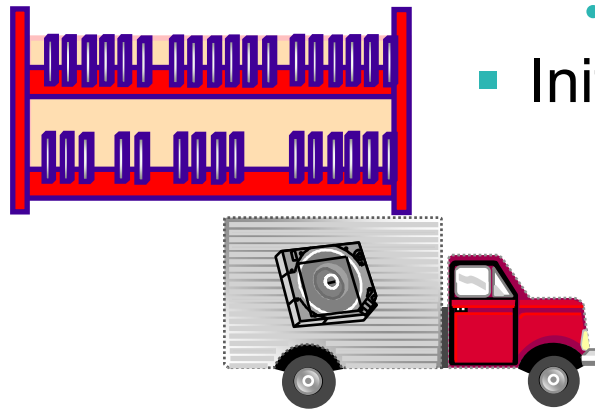
VLPOOL NOTIFY
MEDINF REPLACE

- Apply Retention Limit Controls
- Release (Set Pending Release)
 - Volumes
 - Volume Sets
 - Limit to subsets
- VRS Release Options
 - Ignore EXPDT
 - Scratch Immediate
- Release Actions
 - Notify Owner
 - Replace/INIT/ERASE
- Return Volumes to Scratch
 - Manual Scratch option
 - Limit to subsets



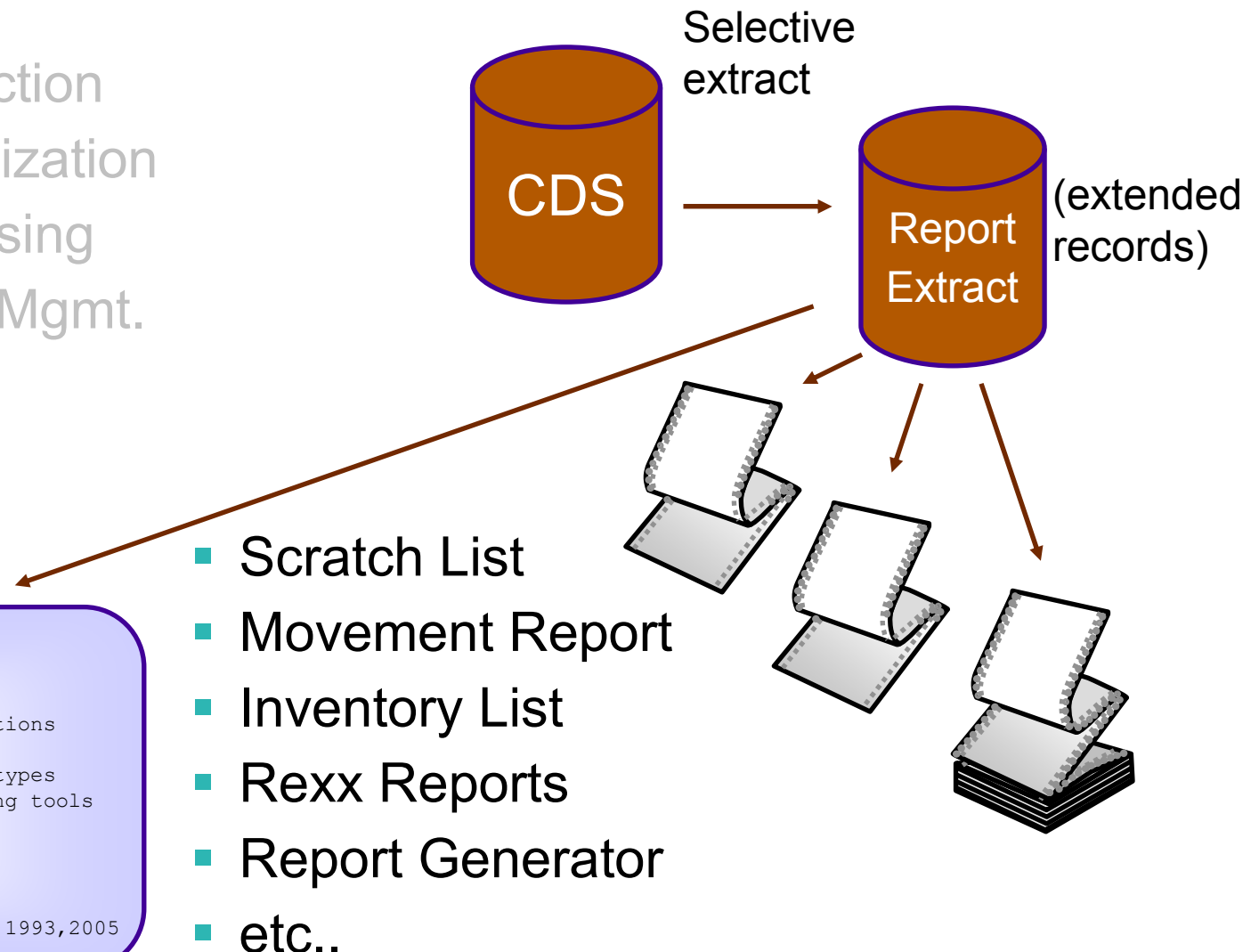
Inventory Management

- Vital Record Selection
 - Catalog Synchronization
 - Expiration Processing
 - **Storage Location Mgmt.**
 - Report Extract
 - Reports
 - Backup
- Assign and release Bins
 - Optionally by Location
 - Reassign started moves
 - Optimized bin Assignment to Minimize slots required
 - Parmlib - REUSEBIN option
 - Initiate/Confirm Moves



Inventory Management

- Vital Record Selection
- Catalog Synchronization
- Expiration Processing
- Storage Location Mgmt.
- **Report Extract**
- **Reports**
- Backup



DFSMSrmm Report Generator
Option ==>

```
0  OPTIONS      - Specify dialog options
1  REPORT      - Work with reports
2  REPORT TYPE - Work with report types
3  REPORTING TOOL - Work with reporting tools
```

Enter selected option or END command.
For more info., enter HELP or PF1.

5695-DF1 (C) COPYRIGHT IBM CORPORATION 1993,2005

Reports

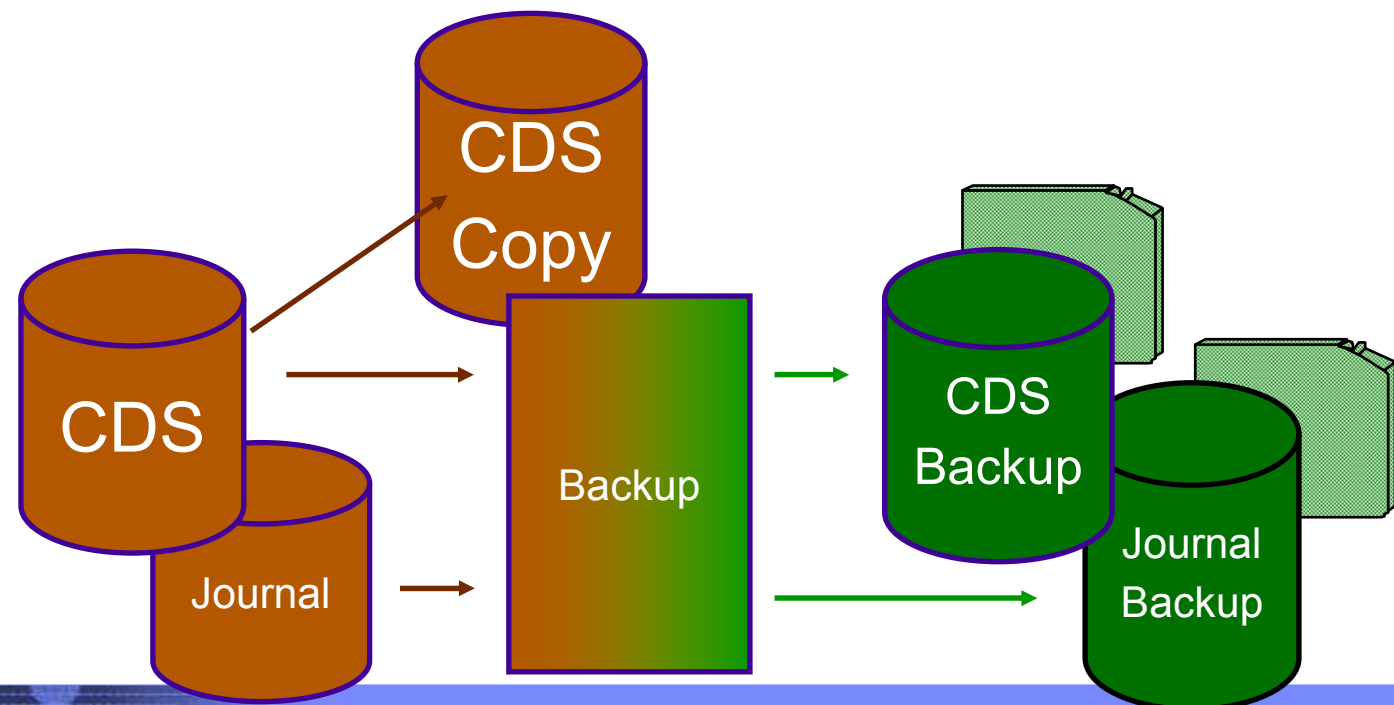
You can obtain information and create reports, using:

- DFSMSrmm ISPF dialog, TSO RMM commands, or the API
- EDGAUD, EDGJRPT JCL, and EDGRPTD report utilities
- EDGRRPTE sample REXX exec to create 17 standard reports
- The DFSMSrmm report generator
- DFSORT™'s ICETOOL utility

Inventory Management

- Vital Record Selection
- Catalog Synchronization
- Expiration Processing
- Storage Location Mgmt.
- Report Extract
- Reports
- Backup

- CDS only
- CDS and Journal
- Journal Only
- Clears Journal



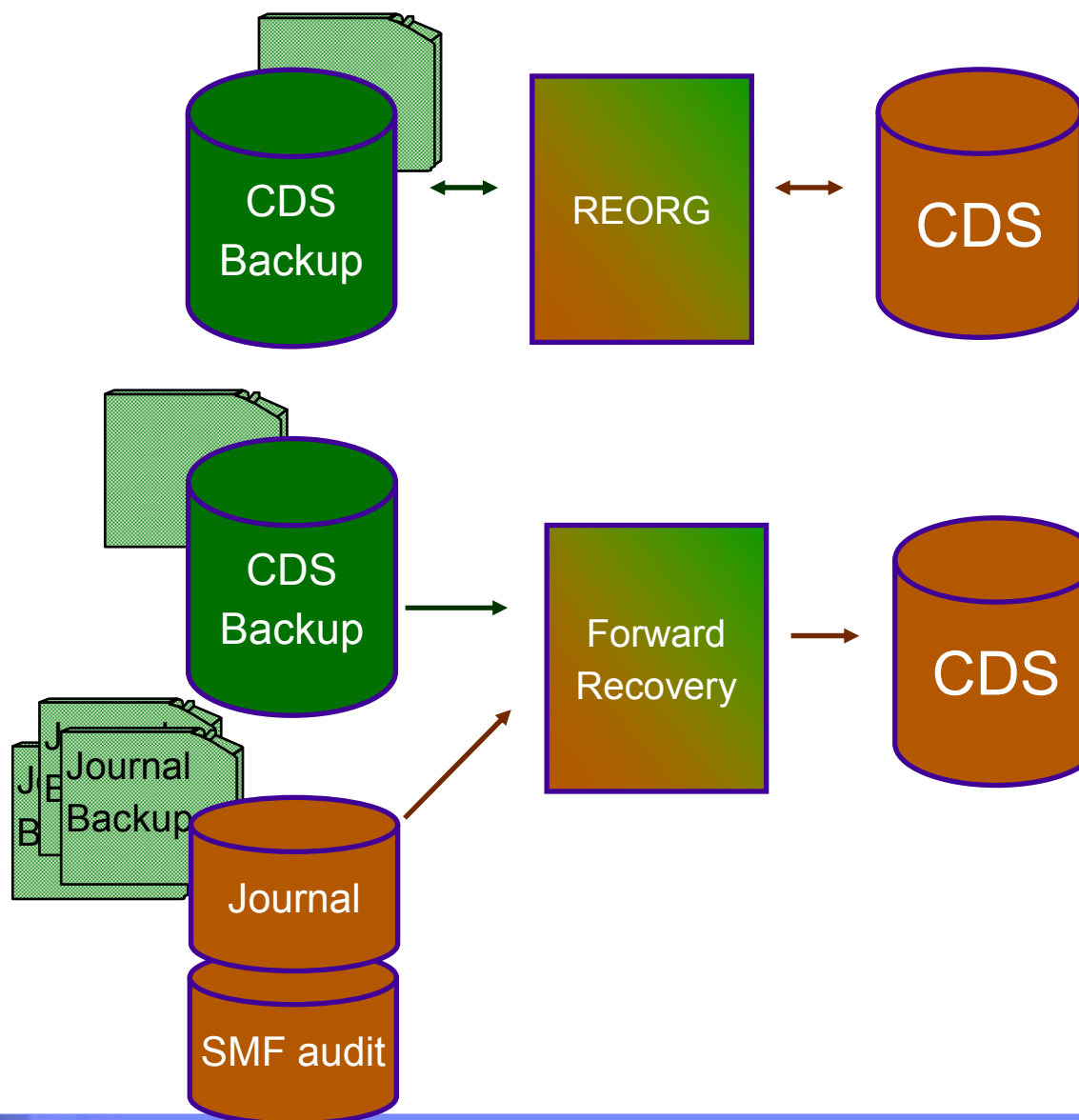
Tools

- Reorg
- Recovery
- Verify / Mend
- Init / Erase
- SMF Reports
- PDA Trace
- IPCS
- Conversion



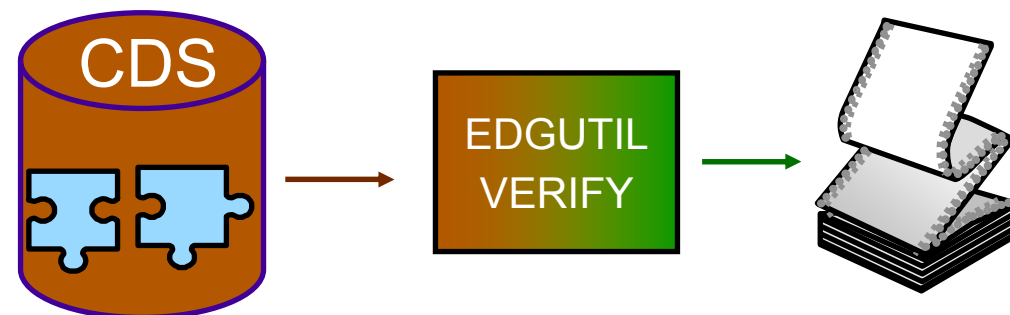
Tools

- **Reorg**
- **Recovery**
- Verify / Mend
- Init / Erase
- SMF Reports
- PDA Trace
- IPCS
- Conversion

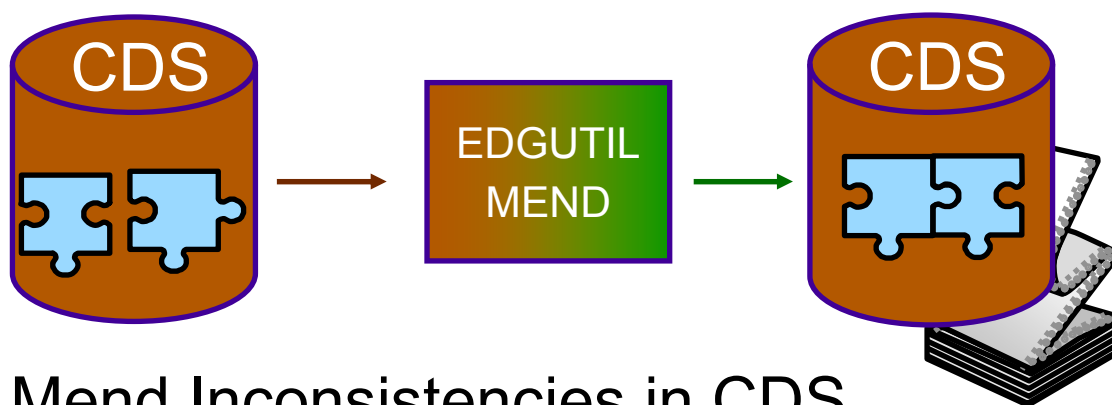


Tools

- Reorg
- Recovery
- **Verify / Mend**
- Init / Erase
- SMF Reports
- PDA Trace
- IPCS
- Conversion



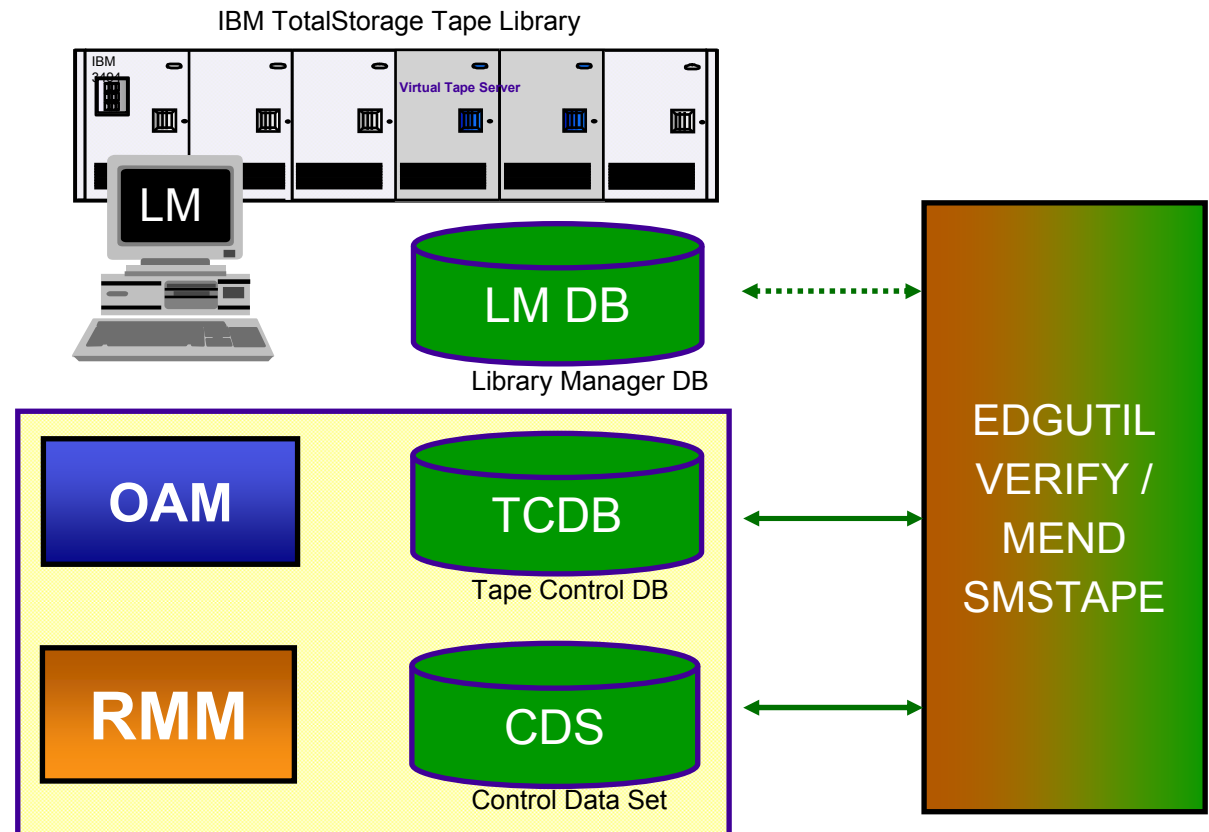
Report Inconsistencies in CDS



Mend Inconsistencies in CDS

Tools

- Reorg
- Recovery
- **Verify / Mend**
- Init / Erase
- SMF Reports
- PDA Trace
- IPCS
- Conversion

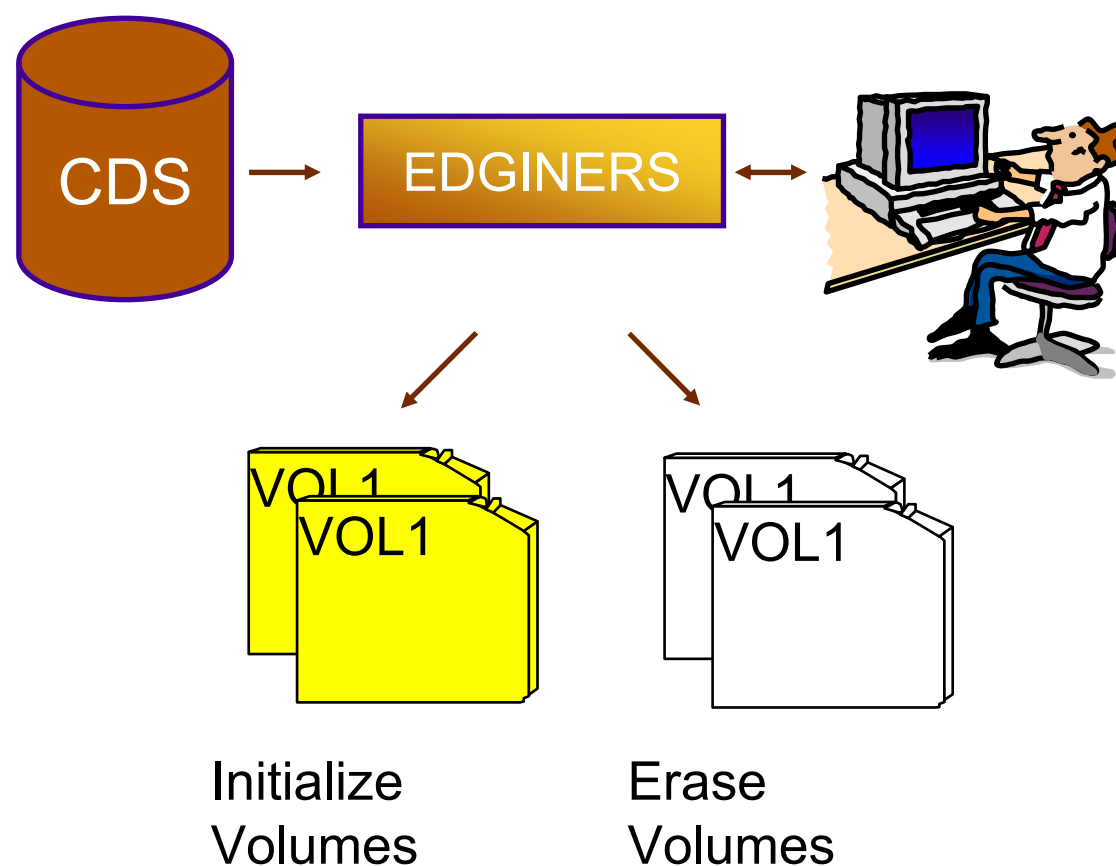


3-Way Audit

Limit to volume and location subsets

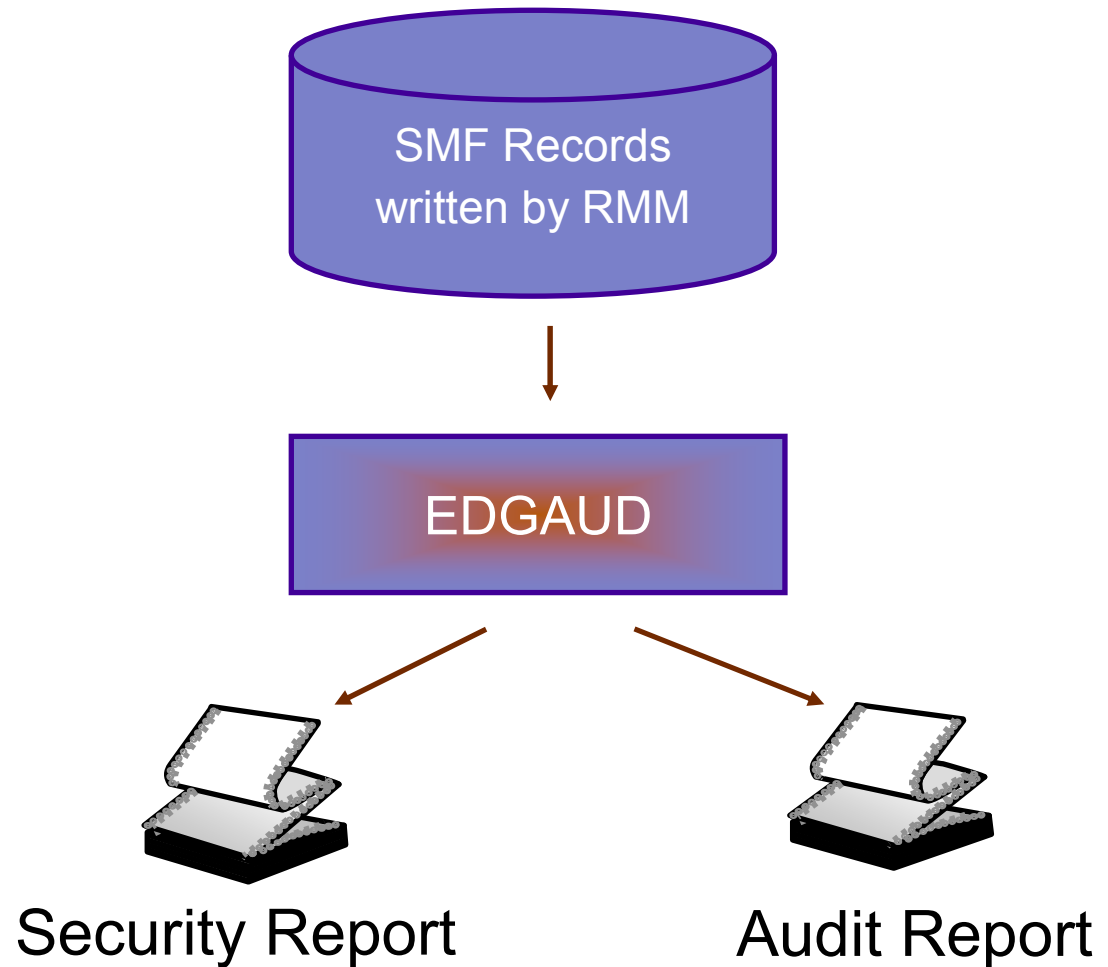
Tools

- Reorg
- Recovery
- Verify / Mend
- **Init / Erase**
- SMF Reports
- PDA Trace
- IPCS
- Conversion



Tools

- Reorg
- Recovery
- Verify / Mend
- Init / Erase
- **SMF Reports**
- PDA Trace
- IPCS
- Conversion



Summary

Protect
Data on Tape

Enforce
Security

Control
Retention

Support
Automation

Manage
Vaulting



Thanks for your interest
in
DFSMSrmm ...

Removable Media Manager